Andrés F. Ramírez-Mejía
BSc, MSc, PhD

Curriculum Vitae
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Education

July 2008 – September 2014	**B.Sc. Biology** University of Caldas (Colombia)
August 2015 – August 2017	**MSc. Conservation and Use of Biodiversity** Pontifical Javeriana University (Colombia)
August 2018 – August 2023	**Ph.D. Biological Sciences** National University of Tucumán (Argentina)
February 2026 – April2027	**MBA Data science** Universidade de São Paulo (Brazil)

Current position

May 2024 - present. Postdoctoral researcher with double affiliation: Department of Ecology, Zoology and Genetics, Institute of Biology, Universidade Federal de Pelotas; Department of Zoology & Physiology, University of Wyoming. *Projects*: Army Ant Followers Project, and Hawaii-VINE Project.

Technical skills

- Experimental design and data collection.
- Causal inference (Pearl framework).
- Data analysis: frequentist and Bayesian statistics (lm, glm, glmm, hierarchical models), Gaussian process models, time series analysis, structural equation modeling, Bayesian networks, network analysis, and simulation studies.
- Machine learning: regression trees, random forests, basic deep learning.
- Programming languages: R (advanced), Python (intermediate) and Stan (intermediate).
- Spatial analysis tools: Google Earth Engine (basic), ArcGis (basic) and R packages: *terra*, *raster*, and *sf* (basic).
- Other tools: Git and GitHub for version control. RMarkdown and Quarto for report generation.

Publication, theses and oral presentations

Published articles

- Harold Pulido-Guarín, Ana Carolina Monmany-Garzia, Alejandra Carla Scannapieco, **Ramírez-Mejía A. F.**, Martín Geria, Alberto Galindo-Cardona. **(2026)**. Three methods for measuring the Earth's magnetic field intensity in Apis mellifera (Hymenoptera: Apidae) drone congregation areas, Environmental Entomology, Volume 55, Issue 3, June 2026, nvag056, https://doi.org/10.1093/ee/nvag056
- **Ramírez-Mejía, A. F.**, Sánchez, F., Urbina-Cardona, J. N., & Ladino, N. **(2026)**. Functional divergence drives the prevalence of low-abundance species in bat assemblages. Functional Ecology, 40(6), 1587-1602. https://doi.org/10.1111/1365-2435.70333
- Wolter, J. H., Vissoto, M., **Ramírez-Mejía, A. F**., Vizentin-Bugoni, J., & Dias, R. A. **(2026)**. Intrinsic and extrinsic drivers shape frugivore functional diversity within a tree population. Ecosphere, 17(2). https://doi.org/10.1002/ecs2.70453
- Gómez-Murillo, L., Vizentin-Bugoni, J., **Ramírez-Mejía, A. F.**, & Tarwater, C. E. **(2026)**. Rainfall alters network structure, while fragmentation results in the breakdown of a mixed-species group of birds. Oecologia, 208(2), 34. https://doi.org/10.1007/s00442-026-05869-7.
- Chacoff, N., Monmany-Garzia, A. C., **Ramírez-Mejía, A. F.**, Castillo, S., Aragón, R. Hacia una agricultura sustentable: ¿Por qué nos importa la biodiversidad? In: Brown, A., & Sánchez, E. Argentina Sustentable: El Norte Grande. 1. ed. Yerba Buena, Argentina: Ediciones Del Subtrópico, 2025. p. 472-476. **ISBN 978-631-90041-4-4**.
- **Ramírez-Mejía A. F.**, Chacoff N., Cavigliasso P., Blendinger P. **(2024)**. How much is enough? Optimizing beehive stocking densities to maximize the production of a pollinator-dependent crop. *Ecological modelling*, vol 498, article number: https://doi.org/10.1016/j.ecolmodel.2024.110891.
- Monasterologo, M, **Ramírez-Mejía A. F.**, et al. **(2024)**. Animal pollination contributes to more than half of Citrus spp. production despite the species and cultivar. *Scientific reports*, vol 14, Article number: 22309. https://www.nature.com/articles/s41598-024-73591-6.

- Fontanarrosa G., Zarbá L., … **Ramírez-Mejía A. F.** … Piquer-Rodríguez M. **(2024)**. Over twenty years of publications in Ecology: Over-contribution of Women reveals a new dimension of gender bias. *PLOS ONE*. https://doi.org/10.1371/journal.pone.0307813
- **Ramírez-Mejía A F**, Chacoff N, Lomáscolo S, Woodcock B, Schmucki R & Blendinger P. **(2024)**. Optimal pollination thresholds to maximize blueberry production. *Agriculture Ecosystems and Environment* https://doi.org/10.1016/j.agee.2024.108903
- **Ramírez-Mejía A F**, Blendinger P, Woodcock B, Schmucki R, Escobar L, Morton R, Vieli L, Nunes-Silva P, Lomáscolo S, Morales C, Murúa M, Agostini K, & Chacoff N. **(2023)**. Landscape structure and farming management interacts to modulate pollination supply and crop production in blueberries. *Journal of Applied Ecology*. DOI: https://doi.org/10.1111/1365-2664.14553
- Nunes-Silva Patrícia, **Ramírez-Mejía Andrés F.**, Blochtein B, Ramos J, Agostini K, Vieli L, Santanna M, Raguse-Quadros M, Maureen M., Chacoff N P, Cavigliasso P, Blendinger P G., Domingos S. **(2023)**. Blueberry: pollination and production in South America. ISBN: 978-65-00-65347-2. DOI: https://doi.org/10.5281/zenodo.7770381.
- **Ramírez-Mejía A F**, Lomáscolo S & Blendinger P. **(2023)** Hummingbirds, honeybees, and wild insect pollinators affect yield and berry quality of blueberries depending on cultivar and farm's spatial context. *Agriculture Ecosystems and Environment*. DOI: https://doi.org/10.1016/j.agee.2022.108229
- Blendinger P G, Rojas T N, **Ramírez-Mejía A F**, Bender I M A, Lomáscolo S, Magro J, Núñez Montellano M G, Ruggera R A, Valoy M & Ordano M. **(2022)** *Nutrient balance and energy-acquisition effectiveness: do birds adjust their fruit diet to achieve intake targets? Functional Ecology*. DOI: doi.org/10.1111/1365-2435.14164
- **Ramírez-Mejía A F**, Urbina-Cardona N, & Sánchez F. **(2022)**. *The interplay of spatial scale and landscape transformation moderates the abundance and intraspecific variation in the ecomorphological traits of a phyllostomid bat*. *Journal of Tropical Ecology*. 38(1), 31-38. DOI: https://10.1017/S026646742100047X
- **Ramírez-Mejía A F**, Echeverry-Galvis M A, & Sánchez F. **(2021)**. *Activity and habitat use by understory birds in a native Andean forest and a eucalypt plantation*. *Wilson Journal of Ornithology*. 132(3): 721-729. DOI: doi.org/10.1676/19-54
- **Ramírez-Mejía A F**, Urbina-Cardona N, & Sánchez F. **(2020)** *Functional diversity of phyllostomid bats in an urban-rural landscape: a scale-dependent analysis*. *Biotropica*. 52(6): 1168-1182. DOI: https://doi.org/10.1111/btp.12816.
- **Ramírez-Mejía A F**, & Sánchez F. **(2016)**. *Activity patterns and habitat use of mammals in an Andean forest and a Eucalyptus reforestation in Colombia*. *Hystrix, the Italian Journal of Mammalogy*, 27(2): 104-110. DOI: https://doi.org/10.4404/hystrix-27.2-11319.
- **Ramírez-Mejía A F**, & Sánchez F. **(2015)**. *Non-volant mammals in a protected area on the Central Andes of Colombia: new records for the Caldas department and the Chinchiná River basin*. *Check List*, 11(2):1-6, Article 1582. DOI: https://dx.doi.org/10.15560/11.2.1582.

Under review

- **Ramírez-Mejía A. F.**, Mary De Aquino, Michael Castaño Díaz, Juliana Hinz Wolter, Henry S. Pollock, J. Patrick Kelley, Jeferson Vizentin-Bugoni & Corey E. Tarwater. Biotic and abiotic factors directly and indirectly impact the structure of non-trophic networks across a rainfall gradient. *Journal: Ecology*. **Role: first author**
- Elizabeth J. Howard, **Andrés F. Ramírez-Mejía**, Michael Castaño, Mary De Aquino, Henry S. Pollock, Jefferson Vizentin-Bugoni, and Corey E. Tarwater. Testing long standing assumptions about army ant swarms: how profitability and dominance alter space use in birds. *Journal: Behavioral Ecology* .**Role: coauthor, data analysis**
- Jordan, Kimberley; **Ramírez-Mejía, Andrés F**.; Wilcox, Rebecca C.; Brawn, Jeffrey ; Alfonso, Camilo; Tarwater, Corey. Testing the space-for-time assumption and the hygric niche hypothesis using precipitation effects on avian body condition. *Journal: Proceedings of the Royal Society B: Biological Sciences*. **Role: coauthor, data analysis**
- **Andrés F. Ramírez-Mejía**, Corey E. Tarwater, J. Patrick Kelley, Jinelle H. Sperry, Jeffrey T. Foster, Donald R. Drake, and Jeferson Vizentin-Bugoni. Temporal dynamics of seed-dispersal networks: Disentangling the role of direct biotic and abiotic factors and bottom-up processes. *Journal: Ecology*. **Role: first author**
- Mary De Aquino1, Henry S. Pollock, J. Patrick Kelley, **Andrés F. Ramírez-Mejía**, Michael Castaño Díaz1, Jeferson Vizentin-Bugoni & Corey E. Tarwater. Species' roles, specialization, and the factors that influence roles in mixed-species animal groups. **Journal of Animal Ecology**. **Role: review**

Code and online datasets

- **Ramírez-Mejía, A.F.**; Blendinger, P.G.; Woodcock, B.A.; Schmucki, R.; Chacoff, N.P. (2024). Wild bee associations with blueberry crops in Argentina, 2020-2021. NERC EDS Environmental Information Data Centre. https://doi.org/10.5285/7756e3fb-0305-48d7-958e-338c02fa33fd
- **Ramírez-Mejía, Andrés F**. et al. (2023). Landscape structure and farming management interacts to modulate pollination supply and crop production in blueberries [Dataset]. **Journal of Applied Ecology**. Dryad. https://doi.org/10.5061/dryad.bg79cnphc

- **Ramírez-Mejía, A F**.; Urbina-Cardona, J N; Sánchez, F (2020). Data from: Functional diversity of phyllostomid bats in an urban-rural landscape: a scale-dependent analysis [Dataset]. **Biotropica**. Dryad. https://doi.org/10.5061/dryad.sn02v6x1p